

Week 2: System Design

1.Game Flow Design

The game begins when both players start at position 0 on the board. Players take turns rolling a virtual dice. Based on the dice value, the player's token moves forward on the board. If a player lands on a ladder, the token moves upward to a higher position. If a player lands on a snake, the token moves downward to a lower position. The process continues until one player reaches position 100 and is declared the winner.

Flow:

Start Game → Roll Dice → Move Player → Check Snake/Ladder → Update Position → Switch Turn → Check Winner → End Game

2. GUI Layout Design

The graphical user interface (GUI) will be designed using Java Swing. The interface will contain:

- 1.Snake and Ladder Board
- 2.Roll Dice Button
- 3.Dice Result Display
- 4.Player 1 Position Label
- 5.Player 2 Position Label
- 6.Current Turn Indicator
- 7.Winner Announcement Area

GUI structure:

SNAKE AND LADDER GAME
GAME BOARD (1-100)
Dice value: 4 Player 1 position : 25 Player 2 position : 18 Current turn: player 1 [Roll Dice] Winner:

3. Class Design

The project will use the following classes:

MainGame

Controls the overall game execution and GUI.

Player

Stores player information such as name and current position.

Dice

Generates random dice values from 1 to 6.

Board

Stores snake and ladder positions and checks player movement.

4. Snake and Ladder mapping:

Start	End
4	25
13	46
33	49
42	63
50	69

Snakes:

Head	Tail
40	3
54	31
66	45
89	53
98	12

5. Use Case Diagram Description

Actor

- Player

Use Cases

- Start Game
- Roll Dice
- Move Token
- Check Snake or Ladder
- View Position
- Win Game

Week 2 Summary

During Week 2, the system architecture and game flow of the Snake and Ladder Game were designed. The graphical user interface layout was planned, required classes were identified, snake and ladder positions were defined, and the overall game structure was prepared to support the development phase in the following weeks.